CS 383

Pumping lemma for context-free languages

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Fall 2026

Recall: Pumping lemma for regular languages

If A is a regular language, then there exists $k \geq 0$ (the pumping length) such that for all strings x, y, z with $xyz \in A$, and $|y| \geq k$, there exist strings u, v, w such that $y = uvw$, $|v| \geq 0$, and for all $i \geq 0$, $xuv^iwz \in A$

We used the contrapositive in the form of the "demon game" to prove that languages are not regular

CF-pumpability (a nonstandard name for this property)

A language L is CF-pumpable with pumping length k if for all $z \in L$ with $|z| \geq k$, there exist strings $u, v, w, x, y \in \Sigma^*$ with $z = uvwxy$ such that

1. $|vx| > 0$; and
2. $|vwx| \leq k$
3. for all $i \geq 0$, $uv^i wx^i y \in L$;

Rather than dividing the string into 3 pieces, we're dividing it into 5

Two of the pieces (v and x) are pumped together

Condition 1 tells us that at least one of v or x must not be ε

Example pumpable language

The language $A = \{w\#w^{\mathcal{R}} \mid w \in \{\mathtt{a}, \mathtt{b}\}^*\}$ is CF-pumpable with pumping length $k = 3$

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Let $u = s$

$v = c$

$w = \#$

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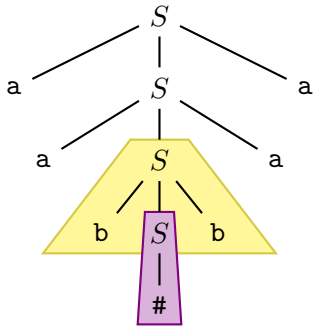
1. $|vx| = |cc| = 2 > 0$
2. $|vwx| = |c\#c| = 3 \leq p$
3. for any $i \geq 0$, $uv^iwx^iy = sc^i\#c^is^{\mathcal{R}} = (sc^i)\#(sc^i)^{\mathcal{R}} \in L$

Parse trees

CFG for $A = \{w\#w^{\mathcal{R}} \mid w \in \{\mathtt{a},\mathtt{b}\}^*\}$: $S \to \mathtt{a}S\mathtt{a} \mid \mathtt{b}S\mathtt{b} \mid \#$

Consider a parse tree for $w = \mathtt{aab\#baa}$

$i = 1$:



$u = \mathtt{aa}$, $v = \mathtt{b}$, $w = \#$, $x = \mathtt{b}$, $y = \mathtt{aa}$

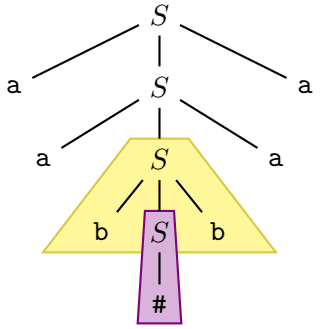
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Parse trees

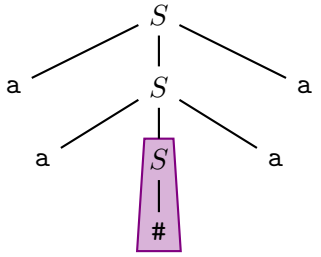
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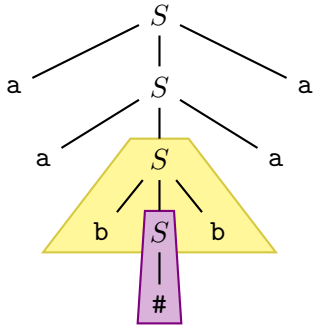
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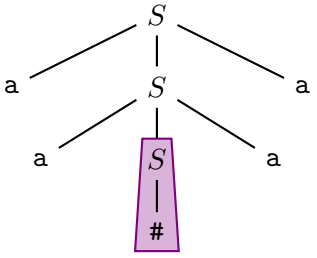
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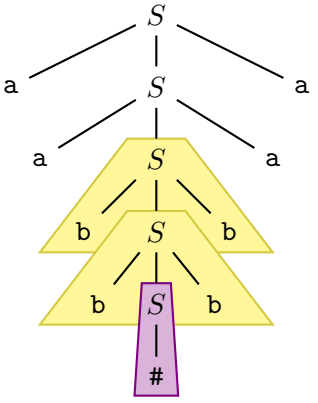
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$i = 2$:



$u = \mathtt{aa}$, $v = \mathtt{b}$, $w = \#$, $x = \mathtt{b}$, $y = \mathtt{aa}$

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CFLs are CF-pumpable

Theorem (Pumping lemma for context-free languages)

Context-free languages are CF-pumpable

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Proof idea.

Consider a CFG $G = (N, \Sigma, P, S)$ in CNF

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Set k large enough that any string of length at least k repeats some nonterminal in its derivation (it turns out $p = 2^{|N|} + 1$ works)

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Theorem (Pumping lemma for context-free languages)

Context-free languages are CF-pumpable

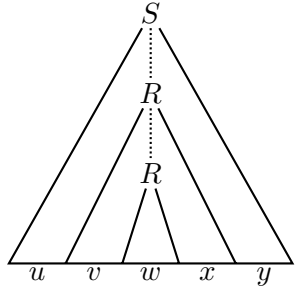
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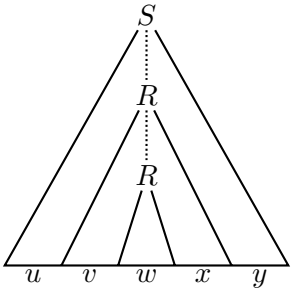
This repeated variable, call it R, will play the same role as the repeated state did in proving that regular languages are pumpable

The diagram to the right shows $S \overset{*}{\Rightarrow} uRy$, $R \overset{*}{\Rightarrow} vRx$, and $R \overset{*}{\Rightarrow} w$ (and thus $R \overset{*}{\Rightarrow} vwx$)



Condition 1: $|vx| > 0$

To see that at least one of v or x is not ε, let's look at $R \overset{*}{\Rightarrow} vRx$

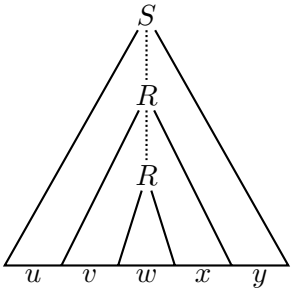


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Since G is in CNF, we must have $R \Rightarrow AB \overset{*}{\Rightarrow} vRx$ for some variables A and B

Two cases:



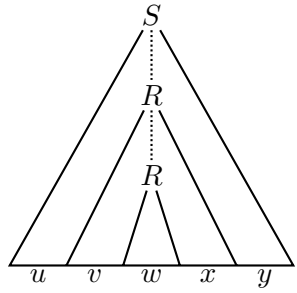
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- $A \overset{*}{\Rightarrow} vRs$ and $B \overset{*}{\Rightarrow} t$ where $st = x$
  t (and thus x) cannot be ε because G is in CNF



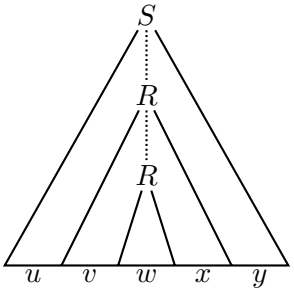
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- $A \overset{*}{\Rightarrow} s$ and $B \overset{*}{\Rightarrow} tRy$ where $st = v$
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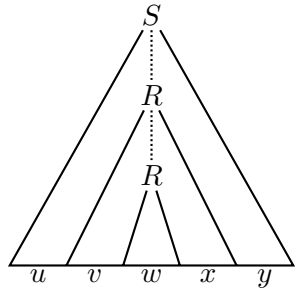
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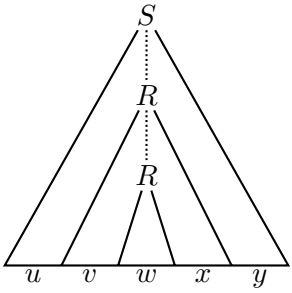
In either case, we've satisfied the first condition:

1. $|vx| > 0$



Condition 2: $|vwx| \leq k$

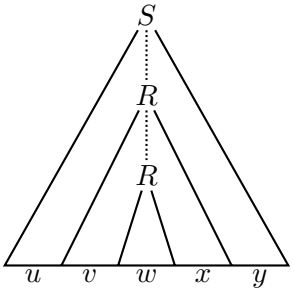
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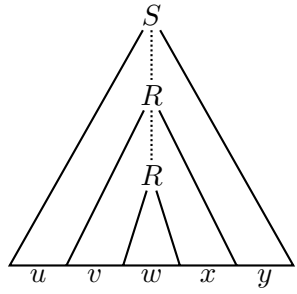
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Now since R is at distance at most $|N| + 1$ from the leaves, we must have $|vwx| \leq 2^{|N|} \leq k$

(A perfect binary tree of height h has 2^h leaves, but the last level of interior nodes in a parse tree for a grammar in CNF have a single child each)



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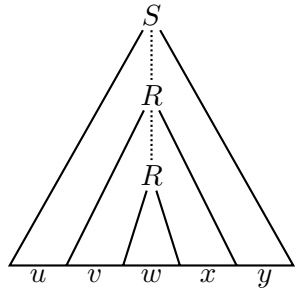
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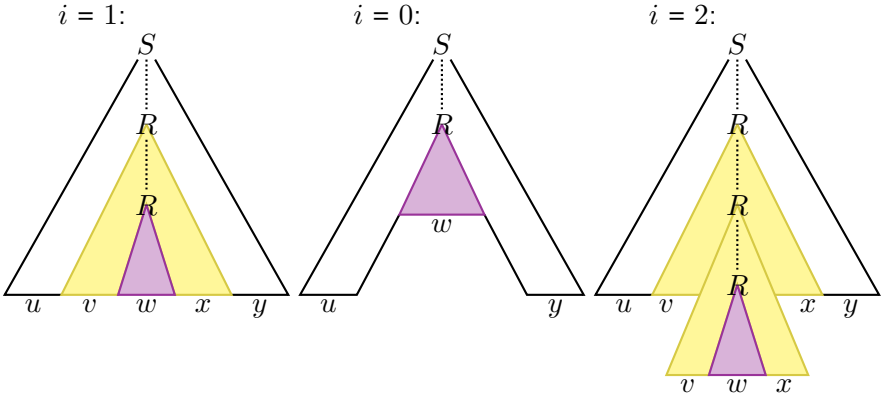
(A perfect binary tree of height h has 2^h leaves, but the last level of interior nodes in a parse tree for a grammar in CNF have a single child each)

Therefore, we've satisfied the second condition:

2. $|vwx| \leq k$



Condition 3: $\forall i \geq 0.\, uv^i wx^i y \in L$



- Pumping down replaces the yellow triangle with the violet triangle
- Pumping up replaces the violet triangle with the yellow triangle
- We can pump up arbitrarily by repeating this process

Thus we've satisfied the final condition:

3. for all $i \geq 0$, $uv^i xy^i z \in L$

Showing that a language is not context-free

We can prove that a language is not context-free by showing that it violates the pumping lemma for context-free languages (proof by contradiction)

Steps:

1. Assume the language, L, is context-free with some unspecified pumping length k
2. Pick string $z \in L$ such that $|z| \geq k$
3. Consider every division of z into $uvwxy = z$ such that $|vx| > 0$, and $|vwx| \leq k$
4. For each possible division, show that for some i, $uv^ixy^iz \notin L$

Demon game version of the contrapositive

To prove a language A is not CF, play (and win!) the following game

1. The demon picks $k \geq 0$
2. You pick $z \in A$ of length at least k
3. The demon picks strings u, v, w, x, y such that $uvwxy = z$, $|vx| > 0$, and $|vwx| \leq k$
4. You pick $i \geq 0$. If $uv^i wx^i y \notin A$, then you win

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$B = \{a^n b^n c^n \mid n \geq 0\}$ is not context-free

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Now consider all possible $uvwxy = z$ with $|vx| > 0$ and $|vwx| \leq k$ (Demon picks u, v, w, x, y)

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$B = \{\mathtt{a}^n\mathtt{b}^n\mathtt{c}^n \mid n \geq 0\}$ is not context-free

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Now consider all possible $uvwxy = z$ with $|vx| > 0$ and $|vwx| \leq k$ (Demon picks u, v, w, x, y)

- At least one of v or x contains two distinct symbols. Then uv^2wx^2y contains symbols out of order so $uv^2wx^2y \notin B$ (You pick $i = 2$)
- Both v and x contain the same symbol ($v = \mathtt{a}^m$, $x = \mathtt{a}^n$; $v = \mathtt{b}^m$, $x = \mathtt{b}^n$; or $v = \mathtt{c}^m$, $x = \mathtt{c}^n$). Then uwz doesn't have the same number of as, bs, and cs, so $uv^0wx^0y \notin B$ (You pick $i = 0$)

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- v and x contain different symbols, but only a single type each ($v = \mathtt{a}^m$, $x = \mathtt{b}^n$; $v = \mathtt{a}^m$, $x = \mathtt{c}^n$; or $v = \mathtt{b}^m$, $x = \mathtt{c}^n$). Again, uwy doesn't have the same number of as, bs, and cs so $uv^0wx^0y \notin B$ (You pick $i = 0$)

Important point about the demon game

Notice that the order of steps matters!

We picked $i = 2$ for the first case but $i = 0$ for the second and third cases

In general, you can pick an i that works for the specific choice of $uvwxy = z$

You still have to consider all possible choices of $uvwxy = z$ though